My Implicit Differentiation AP Practice Problems

All problems are from the AP Calculus website and are the property of the ETS Collegeboard

1998 AP Calculus AB Free-Response Questions

6. Consider the curve defined by $2y^3 + 6x^2y - 12x^2 + 6y = 1$.
   (a) Show that $\frac{dy}{dx} = \frac{4x - 2xy}{x^2 + y^2 + 1}$.
   (b) Write an equation of each horizontal tangent line to the curve.
   (c) The line through the origin with slope -1 is tangent to the curve at point P. Find the x– and y–coordinates of point P.

2000 AP Calculus AB Free-Response Question

5. Consider the curve given by $xy^2 - x^3y = 6$.
   (a) Show that $\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$.
   (b) Find all points on the curve whose x-coordinate is 1, and write an equation for the tangent line at each of these points.
   (c) Find the x-coordinate of each point on the curve where the tangent line is vertical.

2001 AP Calculus AB Free-Response Question

6. The function f is differentiable for all real numbers. The point $\left(3, \frac{1}{4}\right)$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2(6 - 2x)$.
   (a) Find $\frac{d^2y}{dx^2}$ and evaluate it at the point $\left(3, \frac{1}{4}\right)$.

2004 AP Calculus AB Free-Response Question

4. Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

   (a) Show that $\frac{dy}{dx} = \frac{3y - 2x}{8y - 3x}$.

   (b) Show that there is a point P with x-coordinate 3 at which the line tangent to the curve at P is horizontal. Find the y-coordinate of P.

   (c) Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (b). Does the curve have a local maximum, a local minimum, or neither at the point P? Justify your answer.

From CalculusABC.com

The slope of the tangent is -1 at the point (0,1) on $x^3 - 6xy - ky^3 = a$, where k and a are constants. The values of the constants a and k are:

A) $k = 1,\ a = -1$ B) $k = 2,\ a = -2$ C) $k = 3,\ a = -3$

D) $k = -2,\ a = 4$ E) $k = -1,\ a = 3$

Given $x\cos y = x^2 + y^3$, then $\frac{dy}{dx} =$

A) $\frac{2x}{-3y^2 - \sin y}$

B) $\frac{2x}{3y^2 - \sin y}$

C) $\frac{2x + \cos y}{3y^2 - \sin y}$

D) $\frac{2x + \cos y}{3y^2 + x\sin y}$

E) $\frac{2x - \cos y}{3y^2 - x\sin y}$